Solutions

Question 6.1:

(a) $\int x^2 dx = \frac{x^3}{3} + d$

(b) $\int 12x^4 dx = \frac{12x^5}{5} + d$

(c) $\int sin(x) dx = -\cos(x) + d$

(d) $\int cos(x) dx = \sin(x) + d$

(e) $\int e^x dx = e^x + d$

(f) $\int e^{2x} dx = \frac{e^{2x}}{2} + d$

(g) $\int x^2 + 2x + 3 dx = \frac{x^3}{3} + x^2 + 3x + d$

(h) $\int sin(x) + cos(x) dx = -\cos(x) + \sin(x) + d$

(i) $\int e^{2x} + 2x dx = \frac{e^{2x}}{2} + x^2 + d$

(j) $\int e^x + cos(x) dx = e^x + \sin(x) + d$

Question 6.2:

(a) $\int_0^{10} x^2 dx = \left[\frac{x^3}{3}\right]_0^{10} = \frac{10^3}{3} - 0 = 333.33$

(b) $\int_{-10}^{10} x^2 dx = \left[\frac{x^3}{3}\right]_{-10}^{10} = \frac{10^3}{3} - \frac{(-10)^3}{3} = 666.67$

(c) $\int_{-1}^{1} x^3 dx = \left[\frac{x^4}{4}\right]_{-1}^{1} = \frac{1^4}{4} - \frac{(-1)^4}{4} = 0$

(d) $\int_0^{80} \left(-0.1x^2 + 10x\right) dx = \left[-0.1\frac{x^3}{3} + \frac{10x^2}{2}\right]_0^{80} = -0.1\frac{80^3}{3} + 5(80)^2 = 14933.33$

(e) $\int_0^{50} \left(-0.1x^2 + 10x\right) dx = -0.1\frac{50^3}{3} + 5(50)^2 = 8333.33$ (See above)

(f) $\int_0^1 \exp(3x)\, dx = \left[\frac{1}{3}\exp(3x)\right]_0^1 = \frac{1}{3}(\exp(3) - \exp(0)) = 6.3618$

Question 6.3: Find the integrals between the limits specified:

(a) $\int_0^b \frac{h}{b} x dx = \frac{h}{b}\left[\frac{x^2}{2}\right]_0^{10} = \frac{1}{2} \times \frac{h}{b} \times b^2 - 0 = \frac{1}{2}hb$ (half base times height—area of a triangle)

(b) $\int_0^b h dx = h[x]_0^b = h(b-0) = h \times b$ (base times height—area of a rectangle)

(c) $\int_0^T \exp(-\lambda t)\, dt = \left[-\frac{1}{\lambda}\exp(-\lambda t)\right] = -\frac{1}{\lambda}(-\exp(-\lambda t) + \exp(0)) = \frac{1}{\lambda}(1 - \exp(-\lambda T))$

Question 6.4: Use the fact that $\int ax^n = \frac{ax^{n+1}}{n+1} + d$ to integrate the following:

(a) Rewrite $\sqrt{x}$ as $x^{1/2}$ then $\int \sqrt{x} = \frac{2}{3}x^{3/2} + d$

(b) Rewrite $\frac{1}{x^2}$ as x^{-2} then $\int \frac{1}{x^2} = \frac{x^{-1}}{-1} + d = -\frac{1}{x} + d$

(c) Rewrite $\frac{1}{\sqrt{x}}$ as $x^{-1/2}$ then $\int \frac{1}{\sqrt{x}} = 2x^{1/2} + d = 2\sqrt{x} + d$

Question 6.5:We rearrange the decay equation so that all N's are on one side and all t's are on the other and then integrate both sides:

$$\frac{dN}{dt} = \frac{1}{N}$$

We first multiply both sides by N

$$N\frac{dN}{dt} = \frac{\cancel{N}}{\cancel{N}}$$

Then multiply both sides by dt

$$N\frac{dN}{\cancel{dt}} \times \cancel{dt} = dt$$

Then integrate both sides

$$\int N \; dN = \int dt$$

The integral $\int x \; dx$ is equal to $\frac{x^2}{2}$ so we have

$$\frac{N^2}{2} = t + c$$

Rearranging this equation gives us

$$N(t) = \sqrt{2t + c}$$

as the solution to the ODE.

Question 6.6: Perform definite integration of the function to get (and using Question 6.4(b)):

$$\begin{aligned}
\int_{100}^{700} 1000t^{0.5} dt &= 1000 \times \left[t^{3/2}\right]_{100}^{700} \\
&= 1000 * \left(700^{3/2} - 100^{3/2}\right) \\
&= 1.17 \times 10^7 m^3
\end{aligned}$$

Question 6.7: Perform definite integration of the function to get:

$$\begin{aligned}
\int_{3.5}^{6.5} -3t^2 + 30t - 65 dt &= 1000 \times \left[-t^3 + 15t^2 - 65t\right]_{3.5}^{6.5} \\
&= (-(6.5)^3 + 15(6.5)^2 - 65(6.5)) \\
&\qquad -((3.5)^3 + 15(3.5)^2 - 65(3.5)) \\
&= -63.375 + 86.625 \\
&= 23.25
\end{aligned}$$

Question 6.8: We use a technique called *Separation of variables*. We rearrange the decay equation so that all N's are on one side and all t's are on the other and then integrate both sides:

$$\frac{dN}{dt} = -\lambda N$$

We first divide both sides by N

$$\frac{1}{N}\frac{dN}{dt} = -\lambda\frac{\cancel{N}}{\cancel{N}}$$

Then multiply both sides by dt

$$\frac{1}{N}\frac{dN}{\cancel{dt}} \times \cancel{dt} = -\lambda dt$$

Then integrate both sides

$$\int \frac{1}{N}\, dN = -\lambda \int dt$$

The integral $\int \frac{1}{x} dx$ is equal to $\ln x$ so we have

$$\ln N = -\lambda t + c$$

and if we say at $t = 0$, $N = N_0$ we get that $c = \ln N_0$, we therefore obtain

$$N = N_0 \exp(-\lambda t)$$

as the solution to Newton's radioactive equations.

Question 6.9: We want to perform the integral

$$A\int_{t=0}^{t=1} N(t)dt$$

where

$$N = N_0 \exp(-\lambda t)$$

We integrate this equation we follow the steps below:

$$\begin{aligned}
A\int_0^1 N(t)dt &= A\int_0^1 N_0 \exp(-\lambda t)dt \\
&= AN_0\int_0^1 \exp(-\lambda t)dt \\
&= AN_0\left[-\frac{1}{\lambda}\exp(-\lambda t)\right]_0^1 \\
&= \frac{AN_0}{\lambda}\left(-\exp(-\lambda\times 1)+\exp(-\lambda\times 0)\right) \\
&= \frac{AN_0}{\lambda}\left(1-\exp(-\lambda\times 1)\right)
\end{aligned}$$

Plugging in that $t_{1/2} = 30$, $\lambda = \frac{\ln(2)}{t_{1/2}} = \frac{\ln(2)}{30} = 0.0231$, $A = 7.47$, $N_0 = 2.16\times 10^9$ we see that:

$$A\int_0^1 N(t)dt \cong 1.595\times 10^{10}$$

so 1.595×10^{10} atoms we eaten in that time.

Question 6.10: We use a technique called *Separation of variables*. We rearrange the hydrostatic equation so that all P's are on one side and all z's are on the other and then integrate both sides:

$$\frac{dP}{dz} = -\frac{P}{R_a T} g$$

We first divide both sides by P

$$\frac{1}{P}\frac{dP}{dz} = -\frac{1}{R_a T} g$$

Then multiply both sides by dz

$$\frac{1}{P}\frac{dP}{\cancel{dz}} \times \cancel{dz} = -\frac{1}{R_a T} g dz$$

Then integrate both sides

$$\int \frac{1}{P} dP = \int -\frac{g}{R_a T} dz$$

The integral $\int \frac{1}{x} dx$ is equal to $\ln x$ so we have

$$\ln P = -\frac{g}{R_a T} z + c$$

and if we say at $t = 0$, $P = P_0$ we get that $c = \ln P_0$, we therefore obtain

$$P(z) = P_0 \exp\left(-\frac{g}{R_a T} z\right)$$

as the solution to the hydrostatic equation.